

Orthemology Notation Gallery (R7B mathematical-typesetting pipeline, Decision 0023)

DRAFT — not peer reviewed.

Review state (R5): fresh-session repository review completed;

not external human peer review; not empirically validated.

Empirical validation not completed: no designed study has been run.

Terminology benchmark not run: every coined term is a candidate; none adopted.

Companion claim status: philosophical conclusions are conditional on stated premises;

creed-internal material is explicitly school-labeled (Atharī) and revelational where stated.

Source revision (git commit): `daaf87b302a942b77ed98dc89dcf88754ab41614`

SOURCE_DATE_EPOCH: 1784599672 (UTC 2026-07-21, from the source commit)

Repository: theislampill/orthemology

Build: `scripts/build_pdfs.py` (markdown-it-py + typst 0.15.0; deterministic; embedded OFL fonts)

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Orthemology notation gallery

Status: DRAFT — not peer reviewed. Rendered by the R7B mathematical- typesetting pipeline (Decision 0023). Every normative symbol in `docs/notation-registry.yaml` appears below as real mathematics: GitHub renders the `fb0ffb / ``math` source through MathJax, and `scripts/build_pdfs.py` renders `artifacts/notation-gallery.pdf` through the pinned Typst compiler via `scripts/latex_to_typst_math.py`. This is a **typography layer, not a notation redesign**: each row’s meaning is the registry role, unchanged (Decision 0005). `scripts/validate_math_source.py` asserts a bijection between non-retired symbols and this gallery and that `scripts/validate_pdf_math.py` finds no missing-glyph (notdef/tofu) in the rendered PDF — the defect reproduced in `docs/project-closure/r7b/R7B-PDF-MATH-BASELINE.md` (the core  $\mu$ -vector and  $C$ -vector rendering as notdef under the old `#raw` path). Below they compose correctly as $\bar{\mu}$ and $\vec{C}$.

1. Normative symbols rendered as mathematics

Registry key	Rendered	Role
A	A	declared analysis (identifier + version)
alpha, beta, ...	α, β	actor indices
T = task(A)	$T = \text{task}(A)$	task component of the analysis
M_univ, O_univ	$\mathcal{M}, \mathcal{O}$	universal occurrence and ortheme universes
M_A, O_A	$\mathcal{M}_A \subseteq \mathcal{M}, \mathcal{O}_A \subseteq \mathcal{O}$	analysis-active domain and repertoire
Inst_A	$\text{Inst}_A \subseteq \mathcal{M}_A \times \mathcal{O}_A$	analysis-relative instantiation (the primitive)
O*(m; A)	$O^*(m; A)$	true (actual) profile of m under A
Pi_A	Π_A	complete-profile space under A
Pi_A_partial	Π_A^∂	partial-profile space under A
C_hat	$\hat{C}_{A,\alpha,t}(m)$	candidate set of complete profiles
p_hat	$\hat{p}_{A,\alpha,t}(m) \in \Pi_A^\partial$	inferred (partial) profile — belief, never ground truth
Omega	$\Omega_{A,\alpha,t}$	observation map
mu	$\mu = \langle g, S_\mu, \text{select}_\mu, \text{prov}, \text{ver} \rangle$	metaortheme type
pi_mu	π_μ	paired meta-policy
mu_bar	$\bar{\mu}_{e,j}$	the j-th metaorthemma in episode e
MetaInst	$\text{MetaInst}(\bar{\mu}_{e,j}, \mu)$	token typing of a metaorthemma under a metaortheme
Gamma_E	$\Gamma_E = (E, \rightsquigarrow)$	episode DAG
e_Gamma	$e_\Gamma = \text{comp}(\Gamma_E)$	composed boundary-level episode
GoalSchema	$\text{GoalSchema}(\alpha)$	parametric goal schema
G_target	$\mathcal{G}_{\alpha,A_\alpha}$	actor-alpha’s grounded target profile set
Rep_A	Rep_A	representation family under A
chi	$\chi \in \text{Rep}_A$	one representation
L_A_star	$L_A^*(\chi)$	best attainable loss/risk under representation chi
select_mu	select_μ	metaortheme state-selecting evidence procedure
estatus_e	estatus_e	per-claim evidence-status map of episode e
eps_A	ε_A	accepted tolerance of the analysis
theta_stop	θ_{stop}	ANDON stop/escalation risk threshold
K_A	$\mathcal{K}_A$	cause repertoire
R_A	$\mathcal{R}_A$	route repertoire
W_A	$\mathcal{W}_A$	warrant-state repertoire

Q_e	$\mathcal{Q}_e$	claim ledger of episode e
delta_e	δ_e	residual-disposition map of episode e
ReqPath	ReqPath(e)	governance-derived required pathway verdicts

2. Structural constructs (typesetting showcase)

The pipeline handles the full range of publication constructs the corpus uses. These are the constructs named in the Decision 0023 design spike.

2.1 Episode signature (display)

The formal-core episode signature — the exact site of the reproduced notdef defect — rendered as real mathematics ($\backslash\text{vec}\mu$, $\backslash\text{vec } C$ compose):

$$e = \langle \text{id}; m, \kappa, v; x, H; \alpha, w, A, T, t; \bar{\mu}, \text{MetaTok}, \pi; \vec{C}, \hat{p}; r; \text{estatus}; \mathcal{Q}; \delta; \text{hand_in}, \text{hand_out}; a, \text{Succ} \rangle$$

2.2 Multi-line aligned equations

$$\begin{aligned} \text{TokenAdequate}(\bar{\mu}, e) &\Leftrightarrow \text{MetaInst}(\bar{\mu}, \mu) \wedge \text{Compatible}(\bar{\mu}, A(e)) \\ \text{V3c}(e) &\Leftrightarrow \forall \bar{\mu} \in \text{MetaTok}(e) : \text{TokenAdequate}(\bar{\mu}, e) \end{aligned}$$

2.3 Set comprehension and predicate

Grounded target: $\mathcal{G}_{\alpha, A_\alpha} = \{x \in \mathcal{M}_A \mid O^*(x; A_\alpha) \in \Pi_A\}$, and strict soundness
 $\text{StrictlySoundReasoning}_\chi(e) := \text{ReasoningPathAdequate}_\chi(e) \wedge \text{TokenTruthLinked}_\chi(e)$.

2.4 Underbrace

$$\underbrace{O^*(m; A)}_{\text{ground truth, never asserted from discourse}} \neq \underbrace{\hat{p}_{A, \alpha, t}(m)}_{\text{inferred belief}}$$

2.5 Labeled arrows (world transition vs learner update)

$m_t \xrightarrow{a_t} m_{t+1}$ (occurrence lineage) is not $\theta_t \xrightarrow{U_{A(e_t)}} \theta_{t+1}$ (learner update).

2.6 Table containing inline math

Object	Symbol	Non-claim
observation map	$\Omega_{A, \alpha, t}$	not the occurrence
inferred profile	$\hat{p}_{A, \alpha, t}(m)$	not $O^*(m; A)$
metaorthemma	$\bar{\mu}_{e, j}$	not the metaortheme type μ

All notation meanings are those of docs/notation-registry.yaml nothing here introduces a new symbol or claim.